# Functional Requirement Specification (FRS)

## Modular Chilled-Water Cooling Plant with Building Management System Integration

### ISA/IEC 62443 Target Security Level 3 (SL-T 3) — traced from URS-CHW-001

**Template with worked example**

*Worked example: data center plant — Trane Symbio 800 chiller modules · Rockwell ControlLogix 5580 plant control · Siemens Desigo CC (primary) · Kepware historian (secondary, long-term)*

| Field | Value |
|---|---|
| **Document number** | **[FRS-CHW-001]** |
| **Revision** | **[A — issued for review]** |
| **Parent document** | **[URS-CHW-001 Rev A]** |
| **Facility** | **[DC-01 data center, chilled-water plant]** |
| **Asset owner** | **[Owner organization]** |
| **System integrator** | **[Integrator organization]** |
| **Prepared by** | **[Name, role]** |
| **Date** | **[DD Month YYYY]** |
| **Data classification** | **[Business Confidential]** (example marking) |

*Vendor settings and capability statements are as of 28 October 2026; confirm them at FAT, SAT and handover.*

# Using this template

- This FRS turns every functional URS requirement into functions that can be configured, programmed and tested. Each row names the function, the testable requirement, the configuration or programming that delivers it in the worked example, the acceptance criterion and the URS IDs it traces to.
- Black text is the template requirement. *Blue italic text is the worked example.* **[Bracketed blue text]** is a project fill-in with the worked-example default.
- Every functional URS ID (P, Z, S, C, D) traces to at least one FRS row; Annex A proves the coverage. Supplier, lifecycle and verification requirements (URS-L, URS-V) are not functions and are carried to the SDS, HDS and the V&V work instruction.
- Vendor settings are named only where the vendor documents them; anything not confirmed is marked “confirm” and listed in Annex D. Never test a setting on the live plant by changing it — FAT on a representative system, read-only V&V on site.
- This template is open source, with no copyright. The Business Confidential marking is an example of the owner’s data classification; replace it with the owner’s own label.

## Approvals

| Role | Name | Signature | Date |
|---|---|---|---|
| Asset owner — facility operations | **[Name]** | | **[Date]** |
| Asset owner — OT cybersecurity | **[Name]** | | **[Date]** |
| Asset owner — facilities / mechanical engineering | **[Name]** | | **[Date]** |
| System integrator — controls lead | **[Name]** | | **[Date]** |

## Revision history

| Rev | Date | Description | Author |
|---|---|---|---|
| **[A]** | **[Date]** | **[Initial issue for review]** | **[Name]** |

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# 1 Purpose, scope and identifiers

This Functional Requirement Specification (FRS) defines what the modular chilled-water plant control system, its BMS integration and its security functions shall do, in enough detail to configure, program and test them. It answers the URS (URS-CHW-001) requirement by requirement and is the basis for the System Design Specification (SDS), Hardware Design Specification (HDS), FAT, SAT and the V&V, commissioning and acceptance work instruction.

SL-A 3 is achieved by the configured system — zones and conduits, system-level countermeasures, secure configuration and the components' actual capabilities together. Product certification is supporting evidence only.

Scope is the same as URS-CHW-001 section 1. Each FRS row is one testable function and names the URS IDs it satisfies.

## Identifier scheme

| Prefix | Area |
|---|---|
| **FRS-CHL** | Chiller plant control |
| **FRS-HYD** | Chilled- and condenser-water pumping |
| **FRS-CT** | Cooling towers |
| **FRS-ECO** | Waterside economizer |
| **FRS-TES** | Thermal energy storage |
| **FRS-WT** | Water-treatment skid interface |
| **FRS-PRH** | Plant-room HVAC |
| **FRS-MET** | Metering and energy |
| **FRS-MOD** | Operating modes |
| **FRS-DEG** | Degraded modes and fail states |
| **FRS-SAF** | Safety-related interfaces |
| **FRS-HMI** | Operator interface, alarms, trends, historian |
| **FRS-INT** | Interfaces and data exchange |
| **FRS-NET** | Zones, conduits and boundary functions |
| **FRS-SEC-nnn** | System security functions (mirrors URS-S-nnn / SR x.y) |
| **FRS-CMP-nnn** | Component security realization (mirrors URS-C-nnn / CR x.y) |
| **FRS-DEV-xnnn** | Component-type realization (mirrors URS-D-xnnn) |
| **FRS-PLC** | Programming requirements — Secure PLC Coding Top 20 |
| **FRS-LCM** | Maintainability and lifecycle functions |

# 2 References

| Reference | Title | Used for |
|---|---|---|
| [URS-CHW-001 Rev A] | User Requirement Specification — Modular Cooling Plant | Parent requirements |
| ISA/IEC 62443-2-1 | Security program requirements for IACS asset owners | Owner security program |
| ISA/IEC 62443-2-4 | Security program requirements for IACS service providers | Integrator obligations |
| ISA/IEC 62443-3-2 | Security risk assessment for system design | Zones, conduits, SL-T |
| ISA/IEC 62443-3-3 | System security requirements and security levels | Section 7 |
| ISA/IEC 62443-4-1 | Secure product development lifecycle requirements | Supplier SDL |
| ISA/IEC 62443-4-2 | Technical security requirements for IACS components | Section 8 |
| ISA-TR84.00.09 | Cybersecurity related to the functional safety lifecycle | Cyber-informed safety review |
| ANSI/ISA-18.2 | Management of alarm systems for the process industries | Alarm philosophy |
| NIST SP 800-82 Rev. 3 | Guide to OT security | Informative |

| Reference | Title | Used for |
|---|---|---|
| ANSI/ASHRAE 135 | BACnet (incl. BACnet Secure Connect) | BMS protocols |
| ANSI/ASHRAE 15 | Safety standard for refrigeration systems | Refrigerant safety |
| ASHRAE TC 9.9 | Thermal guidelines for data processing environments | Supply temperature basis |
| [Uptime Institute Tier Standard] | Topology (Tier III concurrently maintainable) | Redundancy basis |
| MITRE ATT&CK for ICS | Adversary techniques for ICS | Threat basis (section 4) |
| MITRE EMB3D | Threat model for embedded devices | Device threats (FRS/SDS) |
| Vendor documents | As listed in Annex A for each product | Worked example |
| Secure PLC Coding Top 20 | Top 20 Secure PLC Coding Practices | Section 10 |

# 3 Function allocation

Where each function runs. Zone IDs are as defined in URS section 6.

| Function | Executed by (worked example) | Zone |
|---|---|---|
| Plant sequencing, staging, failover, TES, economizer, setpoint validation | *ControlLogix 5580 plant controllers (A/B per SDS)* | Z2 |
| Chiller capacity control and internal protection | *Trane Symbio 800 (one per chiller)* | Z1-n |
| Pump and fan speed | *PowerFlex 755T; Danfoss FC 102* | Z2 |
| Water chemistry | *Vendor S7-1500 skid* | Z4 |
| Refrigerant detection and emergency ventilation | *Refrigerant monitor panels — hardwired* | None (no network) |
| Plant-room ventilation and heating | *Desigo PXC* | Z10 |
| Supervision, alarms, short-term trends, reports (primary) | *Desigo CC* | Z5 |
| Long-term storage and site / campus aggregation (secondary) | *Kepware with Kepware historian* | Z5 |
| Local plant operation independent of BMS | *Local operator interface (product per SDS)* | Z2 |
| Boundary protection, island mode | *FortiGate (FortiOS 7.6)* | All conduits |
| Identity, time, logging, backup, PKI, monitoring | *OT infrastructure services* | Z7 |

# 4 Process and operational functions

Setpoints, delays and limits are configurable parameters; the values shown are the worked-example defaults the FAT tests against.

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| **Chiller plant control (FRS-CHL)** | | | | | |
| FRS-CHL-001 | **CHW supply temperature control** | The plant controller shall send one common chilled-water leaving setpoint to all running chillers and hold plant supply temperature within ±**[0.5 K]** of setpoint at steady load. BMS setpoint requests shall be accepted only inside the configured range and rate limit. | *Default* **[18.0 °C]**, *range* **[15.0–20.0 °C]**, *rate limit* **[0.5 K/min]**; *validation in an AOI before the value is written to the chillers (FRS-PLC-008).* | FAT simulation and SAT trend: ±0.5 K for 30 min at steady load; out-of-range or over-rate request rejected, alarmed P3 and logged with user. | P-003, P-010, S-305 |
| FRS-CHL-002 | **Chiller start sequence** | Start shall follow: open CHW and CW isolation valves → prove evaporator and condenser flow → enable chiller (network command and hardwired permissive) → confirm running. Any step that times out shall abort the start, raise a P2 alarm and start the next available chiller. | *Valve stroke* **[90 s]**; *flow proof* **[120 s]**; *run proof* **[300 s]** *(confirm against Trane start time).* | Each step time-stamped in the controller log; forced timeout at FAT shows abort and next-chiller start. | P-010, P-015 |
| FRS-CHL-003 | **Staging up** | The plant shall add a chiller when any condition holds for the stage-up delay: supply above setpoint + **[1.0 K]**; any running chiller above **[90 %]** capacity; plant load above **[90 %]** of running capacity. | *Stage-up delay* **[10 min]**; *minimum run time* **[30 min]**. | Simulated load ramp at FAT stages up at the set thresholds and delays. | P-010, P-001 |
| FRS-CHL-004 | **Staging down** | The plant shall remove a chiller when the remaining chillers can carry the load at or below **[70 %]** capacity for the stage-down delay, never dropping below N+1 available units. | *Stage-down delay* **[15 min]**; *minimum off time* **[15 min]**. | Simulated load reduction stages down; redundancy never below N+1 available. | P-010, P-001 |
| FRS-CHL-005 | **Lead / lag rotation** | Lead chiller shall rotate by run hours when the difference exceeds **[168 h]** and the plant is stable; manual rotation shall need the Plant Engineer role. | *Rotation check daily at* **[02:00]**. | Rotation demonstrated; manual rotation refused for Plant Operator role and logged. | P-010, S-201 |
| FRS-CHL-006 | **Duty failure and standby start** | On a duty chiller trip the plant shall command the standby chiller within **[10 s]** of detection and confirm it running within **[60 s]** (subject to the chiller's own restart inhibit), otherwise start the next. | *Trip = Symbio 800 trip status, loss of run feedback or loss of flow.* | FAT: simulated trip → standby command ≤ 10 s; running confirmed ≤ 60 s; TES covers the gap (FRS-TES-001). | P-011, P-002 |
| FRS-CHL-007 | **Demand limiting** | The plant shall accept a chiller demand-limit request within **[40–100 %]** during utility events, validated and rate-limited. | *BMS request through the OPC UA write allowlist (FRS-INT-003).* | Valid request applied; invalid rejected and logged. | P-012, S-305 |
| FRS-CHL-008 | **Chiller communication loss** | If Modbus TCP to a chiller is lost for more than **[10 s]**, the plant shall raise a P2 alarm, mark that chiller unavailable for staging and shall not command it off. The chiller shall keep running under Symbio 800 local control at its local default setpoint. | *Symbio 800 BAS communication-loss behaviour set to continue at local setpoint* **[18 °C]** *(confirm setting name, Annex D).* | FAT: cable pull → P2 alarm, chiller keeps running at local setpoint, staging recalculates. | P-013, S-306, S-701 |
| **Chilled- and condenser-water pumping (FRS-HYD)** | | | | | |
| FRS-HYD-001 | **Variable primary CHW flow** | Running CHW pumps shall share one speed to hold differential pressure at the hydraulically remote point at setpoint. | *DP setpoint* **[100 kPa]** *(range* **[50–150 kPa]**; *PID in plant controller; speed limits* **[30–100 %]**. | DP within ±10 % of setpoint at steady state. | P-010 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-HYD-002 | **Minimum evaporator flow** | The bypass valve shall keep each running chiller at or above its minimum evaporator flow. | *Minimum flow per chiller* **[from Trane selection]**; *bypass valve fail state per FRS-DEG table.* | Simulated low load: flow never below minimum; alarm P2 if bypass at 100 % and flow still low. | P-010, P-024 |
| FRS-HYD-003 | **Pump staging** | Pumps shall stage up when speed exceeds **[90 %]** for **[5 min]** and stage down when n–1 pumps could run below **[50 %]** for **[10 min]**, keeping N+1 available. | | Staging at thresholds in FAT simulation. | P-010, P-001 |
| FRS-HYD-004 | **Pump failover** | On drive fault or no flow proof within **[30 s]**, the standby pump shall start within **[10 s]**. | *Fault = drive fault bit, loss of run feedback or DP collapse.* | FAT: simulated drive fault → standby running within 10 s, P2 alarm. | P-011 |
| FRS-HYD-005 | **Condenser-water pumping** | CW pumps shall start with their chiller and hold condenser flow at setpoint. | *CW flow setpoint* **[per chiller selection]**. | Flow at setpoint ±5 % with chiller running. | P-010 |
| **Cooling towers (FRS-CT)** | | | | | |
| FRS-CT-001 | **CW supply temperature control** | Running tower fans shall share one speed to hold CW supply at setpoint = maximum of **[18 °C]** and wet-bulb + **[3 K]** approach, limited to **[29 °C]**. | *Wet-bulb from plant weather station; fan speed* **[20–100 %]**. | CW supply within ±0.5 K of reset setpoint at steady load. | P-010 |
| FRS-CT-002 | **Cell staging** | Cells shall stage with their isolation valves; a fan shall start only after cell water flow is proven. | *Cell valve proof* **[60 s]**. | Fan start inhibited without flow proof (FAT). | P-010, P-015 |
| FRS-CT-003 | **Fan drive protection** | Fan drive override modes shall be disabled and vibration trips shall be hardwired to the drive. | *FC 102 Fire Mode (par. 24-0*) disabled; vibration switch wired to drive safe-stop input per HDS.* | Parameter readout shows Fire Mode disabled; vibration trip stops fan with no network involvement. | P-053, P-015 |
| FRS-CT-004 | **Freeze protection** | Below **[2 °C]** ambient the plant shall run basin heaters and divert flow through the tower bypass. | | Simulated ambient test at FAT. | P-010 |
| **Waterside economizer (FRS-ECO)** | | | | | |
| FRS-ECO-001 | **Economizer enable / disable** | The economizer shall enable when wet-bulb + approach is at least **[1 K]** below CHW return for **[30 min]** and disable when that margin is lost for **[10 min]**. | *Approach* **[3 K]**. | Enable and disable at thresholds in simulation. | P-005 |
| FRS-ECO-002 | **Partial and full economizer** | The heat exchanger shall pre-cool CHW return upstream of the chillers; chillers shall stage off only when the HX alone holds supply setpoint. | | Supply stays within ±0.5 K during partial operation. | P-005, P-010 |
| FRS-ECO-003 | **Bumpless changeover** | Economizer transitions shall not move CHW supply by more than **[1 K]**. | *Valve ramp* **[5 min]**. | Trend shows ≤ 1 K excursion. | P-005, P-003 |
| **Thermal energy storage (FRS-TES)** | | | | | |
| FRS-TES-001 | **Ride-through discharge** | The TES discharge valve shall open automatically when CHW supply exceeds setpoint + **[1 K]** with insufficient chiller capacity, or on a power-interruption signal, and hold supply within setpoint + **[2 K]** for **[10 min]** at full load. | *Discharge valve fails open on loss of control power (FRS-DEG table).* | SAT: simulated chiller loss → valve open ≤ 10 s; trend meets P-004. | P-004, S-306 |
| FRS-TES-002 | **Recharge** | After an event the tank shall recharge when spare chiller capacity is at least **[20 %]**, completing within **[4 h]**. | | Recharge completes in time on a test event. | P-004 |
| FRS-TES-003 | **Tank state monitoring** | Tank charge shall be calculated from stratification sensors; below **[80 %]** outside an event raises a P2 alarm. | | Alarm on simulated low charge. | P-004, P-031 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| **Water-treatment skid interface (FRS-WT)** | | | | | |
| FRS-WT-001 | **Skid autonomy and status** | The vendor skid shall control water chemistry on its own S7-1500. The plant controller shall receive status over Modbus TCP and a hardwired summary trip, and shall not write to the skid. | *S7-1500 Modbus server data area exposed read-only to the plant controller (method per Siemens MB_SERVER documentation — confirm); no write registers mapped in the MVI56E-MNET.* | Write attempt from the plant side is refused (FAT); summary trip works with the network disconnected. | P-012, S-201, S-305 |
| FRS-WT-002 | **Skid alarms and data to BMS** | Skid alarms and values shall reach Desigo CC (primary) and Kepware (secondary) over OPC UA SignAndEncrypt. | *S7-1500 OPC UA server: guest access off, user or certificate authentication, two client certificates trusted.* | Alarms appear in Desigo CC within **[5 s]**; values logged in Kepware historian. | P-031, S-301 |
| **Plant-room HVAC (FRS-PRH)** | | | | | |
| FRS-PRH-001 | **Ventilation and heating** | Desigo PXC shall hold plant-room temperature below **[30 °C]** with supply/exhaust fans and above **[10 °C]** with unit heaters. Refrigerant emergency ventilation is hardwired and independent (FRS-SAF-001). | *Desigo PXC programs; BACnet MS/TP to room controllers.* | Functional test of both modes; PXC cannot inhibit emergency ventilation. | P-050, P-031 |
| **Metering and energy (FRS-MET)** | | | | | |
| FRS-MET-001 | **Electrical metering** | kW, kWh, PF, V, A and THD shall be read from every chiller feeder and MCC feeder meter, read-only. | *PM8000 (BACnet/IP) and PM5560 (Modbus TCP) polled by Desigo CC every* **[15 s]** *and by Kepware every* **[5 s]**. | All points present and plausible against a clamp-meter spot check. | P-040, S-301 |
| FRS-MET-002 | **Thermal load** | Plant thermal load shall be calculated in the plant controller as flow × ΔT × fluid properties every **[1 s]**, filtered over **[60 s]**. | *Promag flow + matched supply/return RTDs.* | Within ±**[5 %]** of a reference calculation at SAT. | P-041, P-010 |
| FRS-MET-003 | **Plant efficiency** | Plant efficiency (kW electrical per kW cooling) shall be calculated every **[1 min]**, trended and reported daily. | *Calculated in Desigo CC; stored long-term in Kepware historian.* | Report generated and matches hand calculation for a sample hour. | P-040 |
| **Operating modes (FRS-MOD)** | | | | | |
| FRS-MOD-001 | **Plant modes** | Modes Auto, Manual, Maintenance lockout, Economizer (automatic sub-mode) and Emergency shall exist; changes shall need the role in the mode matrix and be logged with user, time, old and new value. | *Mode matrix: Plant Operator — none; Shift Supervisor — Emergency; Plant Engineer — Manual, Maintenance lockout.* | Unauthorized role refused; audit record complete. | P-014, S-201, S-212 |
| FRS-MOD-002 | **Emergency mode** | Emergency mode shall start automatically when CHW supply exceeds setpoint + **[3 K]** for **[2 min]**, or manually: all available chillers and pumps run, TES discharges, economizer locks out if adverse. | | Automatic entry and exit demonstrated at FAT. | P-014, P-011 |
| FRS-MOD-003 | **Maintenance lockout** | A unit in maintenance lockout shall leave rotation and staging; physical lockout-tagout remains the energy-isolation control. | | Locked-out unit never commanded; audit record present. | P-014, P-002 |
| FRS-MOD-004 | **Manual control timeout** | Manual equipment commands shall revert to Auto after **[8 h]** unless renewed. | | Timeout demonstrated. | P-014 |
| **Degraded modes (FRS-DEG)** | | | | | |
| FRS-DEG-001 | **Loss of BMS** | Plant control shall continue in its current mode holding the last validated setpoints; loss of the Desigo CC OPC UA session for **[60 s]** raises an alarm on the local operator interface. | *OPC UA session watchdog tag in the controller.* | SAT: BMS servers stopped → cooling unaffected, local alarm. | P-020, S-306 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-DEG-002 | **Loss of IDMZ, enterprise or internet** | Plant control and local operation shall be unaffected; Kepware historian keeps storing locally and forwards to campus aggregation when the link returns. | | SAT: C7 and C12 opened → no plant effect; campus backlog forwarded after restore. | P-021, S-501 |
| FRS-DEG-003 | **Loss of plant controller** | Loss of one plant controller (A/B) shall not interrupt cooling; loss of both shall leave chillers under local control, drives at their configured communication-loss action and valves in their fail states. | *Architecture per SDS; fail states per table below.* | FAT: controller stop tests for A, B and both. | P-002, P-013, S-306 |
| FRS-DEG-004 | **Network single failure** | A single link or switch failure shall not cause loss of control of more than one redundant unit; ring recovery within **[100 ms]**. | *DLR or PRP per SDS on Stratix 5800.* | FAT: link and switch pulls; no loss of control. | P-022, S-701 |
| FRS-DEG-005 | **Power interruption and restart** | Controls, network and servers shall ride through on UPS for **[15 min]**; after power returns the plant shall restart automatically to the pre-event mode, with chiller starts staggered by **[60 s]**. | *Retentive mode and setpoint tags; restart logic per FRS-PLC-015.* | SAT: simulated power interruption and restoration. | P-023, S-705 |
| FRS-DEG-006 | **Defined fail states** | Every final element shall go to the state in the fail-state table on communication loss, controller fault and control-power loss. | *See table.* | Each element tested for each condition at FAT. | P-024, S-306 |
| **Safety-related interfaces (FRS-SAF)** | | | | | |
| FRS-SAF-001 | **Refrigerant detection and emergency ventilation** | Refrigerant monitor alarm shall start emergency ventilation and alarms through hardwired relays per **[ANSI/ASHRAE 15 and local code]**; the BMS shall only monitor (QuickServer, read-only). | *No network write path to the monitor panel.* | Functional test with calibration gas; works with all networks disconnected. | P-050, Z-004 |
| FRS-SAF-002 | **Chiller internal protections** | No chiller protection setpoint shall be mapped to any writable network point. | *Symbio 800 integration map reviewed; protections not in the Modbus or BACnet write list.* | Map review and FAT write attempts refused. | P-051 |
| FRS-SAF-003 | **Fire alarm shutdown** | Fire alarm shutdown signals shall be hardwired to the plant. | | Functional test with networks disconnected. | P-052, Z-004 |
| FRS-SAF-004 | **Protection-defeating functions** | Override, bypass and force functions shall be disabled or restricted to an authorized role and their state alarmed. | *FC 102 Fire Mode disabled; ControlLogix forces disabled and Forces Enabled status alarmed P2; S7-1500 force jobs alarmed by the skid.* | Readout and alarm test. | P-053, S-304 |
| FRS-SAF-005 | **Cyber-informed safety review** | An ISA-TR84.00.09 review shall show that no network path can disable a safety protection. | *Review record covers SAF-001 to SAF-004.* | Review approved by owner before SAT. | P-054 |
| FRS-SAF-006 | **Hardwired permissives** | Flow proving, condenser flow and emergency stop shall be in the hardwired enable circuit of each chiller and pump. | *Per HDS wiring.* | Permissive test with networks disconnected. | P-015, Z-004 |
| **Maintainability and lifecycle (FRS-LCM)** | | | | | |
| FRS-LCM-001 | **Supportability record** | Each component's security support end date shall be recorded in the inventory. | *Inventory field per URS-S-708.* | Inventory complete at handover. | P-060, S-708 |
| FRS-LCM-002 | **Spares at baseline** | Spares shall be stored at the approved firmware and configuration baseline with a hash record. | | Spot check of two spares at SAT. | P-061 |
| FRS-LCM-003 | **Management of change** | Every control or security change after SAT shall carry an MOC number in the controller, BMS or firewall change comment. | *Change comment field used in Studio 5000, Desigo CC and FortiGate.* | Audit sample shows MOC numbers. | P-062, S-212 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-LCM-004 | **Engineering from Z6 only** | Engineering connections to Z1–Z4 and Z10 shall be accepted only from engineering workstations in Z6. | *Conduit C3 rules; CIP Security policy; Symbio UI and meter web access restricted to Z6 addresses.* | Connection attempt from Z5 refused. | P-063, S-201 |

## 4.1 Fail-state table (FRS-DEG-006)

| Final element | Communication loss | Controller stop or fault | Control power loss | Configured in |
|---|---|---|---|---|
| **Chiller enable (Symbio 800)** | Continue at local setpoint | Continue at local setpoint | Chiller stops; restarts per Trane restart logic | *Symbio 800* |
| **CHW pump drive (PowerFlex 755T)** | Hold last speed | Hold last speed | Pump stops; restarts on power return | *Drive communication-loss action* |
| **CW pump drive (PowerFlex 755T)** | Hold last speed | Hold last speed | Pump stops; restarts on power return | *Drive communication-loss action* |
| **Tower fan drive (FC 102)** | Freeze output | Freeze output | Fan stops | *Par. 8-03 timeout* **[10 s]**, *8-04 Freeze output* |
| **Chiller isolation valves** | Hold position | Hold position | Fail in place | *Actuator selection (HDS)* |
| **Minimum-flow bypass valve** | Hold position | Hold position | Fail open | *Actuator selection (HDS)* |
| **Economizer HX valves** | Hold position | Hold position | Fail closed (HX isolated) | *Actuator selection (HDS)* |
| **TES discharge valve** | Hold position | Open | Fail open | *Spring-return actuator (HDS)* |
| **Tower bypass valve** | Hold position | Hold position | Fail to tower | *Actuator selection (HDS)* |

# 5 Operator interface, alarms, trends and historian

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-HMI-001 | **Local plant operator interface** | A local operator interface in Z2 shall show plant overview, equipment status and alarms and allow authorized control independent of the BMS. | *Product per SDS; login per FRS-SEC-101.* | SAT with Desigo CC servers stopped. | P-030, P-020 |
| FRS-HMI-002 | **BMS graphics and control** | Desigo CC shall present plant, equipment and alarm graphics; control actions limited to the OPC UA write allowlist (FRS-INT-003). | *Desigo CC graphics per owner standard.* | Graphic walk-through; non-allowlisted command not possible. | P-031, P-012 |
| FRS-HMI-003 | **Alarm priorities** | Alarms shall use four priorities per the owner alarm philosophy ([ANSI/ISA-18.2]). | *P1 imminent loss of cooling; P2 equipment trip or loss of redundancy; P3 degraded or security event; P4 maintenance. Examples in the alarm matrix.* | Alarm list reviewed against matrix; P1 audible. | P-031 |
| FRS-HMI-004 | **Alarm attributes, shelving and suppression** | Each alarm shall have tag, priority, setpoint, deadband and delay; shelving shall need Shift Supervisor role, a reason and a timeout of at most [8 h]. | | Shelving refused for Plant Operator; auto-unshelve demonstrated. | P-031, S-201, S-304 |
| FRS-HMI-005 | **Security-relevant alarms** | Events that affect operation shall alarm to operators: controller mode or keyswitch change, logic change detected, firmware change, communication loss, boundary island mode active, NTP loss, log collector down. | *Sources: Logix change detection and controller status; FortiGate automation; NTP and collector health.* | Each event triggered at FAT and seen at the operator console. | P-033, S-304, S-210 |
| FRS-HMI-006 | **Short-term trends** | Desigo CC shall trend key analog values at [1 min] and digital values on change, retained [13 months]. | | Trend retention configuration readout. | P-032 |
| FRS-HMI-007 | **Long-term historian** | Kepware historian shall be the long-term store: all plant tags at [1 s] (critical) and [10 s] (others), retained [7 years], read-only for users; it aggregates site and campus data outbound through the IDMZ. | *Kepware OPC UA and BACnet/Modbus clients read-only; storage on encrypted volume in Z5; forwarding over C12.* | Retention, sample rate and read-only access verified; outbound-only forwarding. | P-032, S-401, S-501 |
| FRS-HMI-008 | **Reports** | Daily plant performance, monthly energy and weekly security-event reports shall be generated automatically. | *Desigo CC reports; security report from the Z7 log collector.* | Reports produced for one test week. | P-040, S-601 |

## 5.1 Alarm priority matrix (worked example)

| Priority | Examples | Operator response |
|---|---|---|
| **P1 — Critical** | *CHW supply > setpoint + 3 K; all chillers off; TES < 20 % during event; both plant controllers faulted* | Immediate |
| **P2 — High** | *Chiller, pump or fan trip; loss of N+1; chiller communication loss; forces enabled* | ≤ 5 min |
| **P3 — Medium** | *Economizer fault; setpoint request rejected; controller mode change; logic change detected; island mode; NTP or log collector loss* | ≤ 30 min |
| **P4 — Low** | *Maintenance due; filter differential; runtime thresholds* | Next shift |

# 6 Interfaces and data exchange

Desigo CC is the primary supervisory path for every subsystem; Kepware is the secondary path and its historian is the long-term site and campus-wide store. Write access exists only where a row lists it.

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-INT-001 | **Symbio 800 ↔ plant controller (C1-n)** | Modbus TCP: MVI56E-MNET client polls each Symbio 800 every **[1 s]**; communication fail after **[10 s]**. Writes only: chiller enable, CHW leaving setpoint (validated), demand limit (validated). | *Register map from the Trane Symbio 800 Modbus integration guide (confirm, Annex D). All other registers read-only in the client map.* | FAT: write to a non-listed register is impossible from the map; FortiGate log shows only the plant controller on TCP 502. | P-012, S-102, S-305, Z-003 |
| FRS-INT-002 | **Symbio 800 ↔ Desigo CC (C9-n)** | BACnet/SC when Trane supports it; until then BACnet/IP read-only. Desigo CC shall not write to chillers. | *Desigo CC chiller objects imported read-only; FortiGate blocks BACnet write services on C9-n where its industrial signatures support it (confirm).* | Write attempt from Desigo CC refused or blocked and logged. | P-012, S-301, Z-005 |
| FRS-INT-003 | **ControlLogix OPC UA server ↔ Desigo CC (C2, primary)** | OPC UA on **[4840]**, SignAndEncrypt, security policy Basic256Sha256 or stronger, X.509 application certificates from the owner PKI, user authentication; anonymous disabled. Exposed tags limited to the BMS interface set; writes limited to the allowlist: plant mode request, CHW setpoint request, demand-limit request, alarm acknowledge. | *Controller-native OPC UA server configured in Studio 5000 per 1756-UM543 (firmware revision and limits: Annex D); tags outside the BMS interface set have External Access = None or Read Only.* | FAT: SignAndEncrypt only endpoint offered; write to non-allowlisted tag refused; certificate from untrusted CA refused. | P-012, S-102, S-301, S-403, Z-005 |
| FRS-INT-004 | **ControlLogix OPC UA server ↔ Kepware (C2, secondary)** | Kepware OPC UA client, same security mode, read-only user role; sample rate **[1 s]** critical, **[10 s]** others; session count kept inside controller limits. | *Separate OPC UA user for Kepware with read-only rights.* | Kepware cannot write any tag; controller session count within limits under load. | P-032, S-201, S-702 |
| FRS-INT-005 | **S7-1500 skid ↔ Desigo CC and Kepware (C10)** | OPC UA SignAndEncrypt; Desigo CC primary (alarms and values), Kepware secondary (read-only). | *S7-1500 OPC UA server: guest off, user or certificate authentication, server certificate from owner PKI.* | Both clients connect with SignAndEncrypt only; Kepware write refused. | P-031, S-301, Z-005 |
| FRS-INT-006 | **S7-1500 skid ↔ plant controller (C5)** | Modbus TCP status read by the MVI56E-MNET every **[2 s]**; plus hardwired summary trip. | *Read-only data area (FRS-WT-001).* | Status within 2 s; trip independent of network. | P-012, Z-003 |
| FRS-INT-007 | **Refrigerant and leak panels ↔ Desigo CC (Z4, C10)** | Modbus RTU to the QuickServer; QuickServer serves BACnet/IP to Desigo CC, read-only. | *QuickServer map has no write-capable points; HTTPS mode for FS-GUI.* | Desigo CC write refused; FS-GUI only on HTTPS. | P-050, S-301, Z-005 |
| FRS-INT-008 | **Desigo PXC ↔ Desigo CC (C11)** | BACnet/SC with owner-PKI certificates; BACnet MS/TP below the PXC only. | *PXC as BACnet/SC node to the Desigo CC hub.* | Packet capture on C11 shows TLS only. | S-301, Z-005 |
| FRS-INT-009 | **Meters ↔ Desigo CC and Kepware (C4)** | PM8000 BACnet/IP and PM5560 Modbus TCP, read-only to both clients. | *PM5560 Modbus IP filtering enabled with the two client addresses; PM8000 BACnet objects read-only.* | Write from a third address refused or blocked. | P-040, S-102, Z-003 |
| FRS-INT-010 | **Drives and flowmeters ↔ plant controller (Z2)** | CIP EtherNet/IP; CIP Security on PowerFlex 755T; FC 102 (MCA 121) and Proline 300/500 without CIP Security unless confirmed. Requested packet interval **[100 ms]**. | *FactoryTalk Policy Manager zones for CIP Security devices.* | Connection faults alarmed P2; CIP Security policy deployed and verified for 755T. | S-102, S-301, Z-005 |
| FRS-INT-011 | **Kepware historian ↔ campus aggregation (C12)** | Outbound-only forwarding from Kepware through the IDMZ to the campus historian aggregation; no inbound connection to Z5. | *Protocol per SDS (OPC UA or HTTPS); FortiGate allows Z5→Z8 only.* | Inbound attempt from Z8 blocked. | P-021, S-113, S-502 |
| FRS-INT-012 | **Hardwired signals** | Chiller and pump enable permissives, flow proving, emergency stop, refrigerant emergency ventilation, fire alarm shutdown and skid summary trip shall be hardwired. | *Per HDS I/O list.* | Each signal functionally tested with networks disconnected. | P-015, P-050, P-052, Z-004 |

# 7 Zones, conduits and boundary functions

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-NET-001 | **Zone implementation** | Each zone shall be a separate subnet on its own switch set (or dedicated switch ports) and a dedicated FortiGate interface; no host shall have interfaces in two OT zones except the boundary devices. | *VLAN and subnet plan per SDS.* | Network scan from each zone at FAT matches the plan. | Z-001, Z-002, S-501 |
| FRS-NET-002 | **Conduit rule register** | Every conduit shall be implemented as deny-all plus listed rules (source, destination, protocol, port, direction, justification, owner, review date); no any-any rules. | *FortiGate policy per conduit; rule register delivered per URS-L-011.* | Rule review: every rule maps to a conduit row; no any-any. | Z-003, S-502 |
| FRS-NET-003 | **Protocol-aware filtering** | Where the boundary device can inspect the protocol, write services shall be limited to listed sources and points: Modbus function codes 5, 6, 15 and 16 only from the plant controller on C1-n; BACnet write services blocked on C4 and C9-n. | *FortiGate industrial application control signatures (confirm coverage, Annex D).* | FAT: blocked writes logged. | Z-005, S-305, S-502 |
| FRS-NET-004 | **Island mode** | The Security Administrator shall be able to isolate any zone from all others in one action; cooling shall continue. | *Pre-staged FortiGate policy set or automation stitch per zone.* | SAT: island Z5 and Z7 → cooling unaffected, P3 alarm on local interface. | S-502, P-020 |
| FRS-NET-005 | **Fail close** | Loss of both boundary units shall block inter-zone traffic while intra-zone control continues. | *FortiGate HA pair; no bypass ports on OT conduits.* | FAT: power both units off → conduits closed, plant runs. | S-502 |
| FRS-NET-006 | **Secure protocol selection** | Each conduit shall use the secure protocol variant where both ends support it; every insecure protocol use shall be listed with its compensating countermeasure. | *Protocol table (URS section 6.4).* | Packet capture per conduit matches the table. | Z-005, S-301 |
| FRS-NET-007 | **Plant control network resilience** | The Z2 network shall use a ring or parallel-redundancy topology. | *DLR or PRP on Stratix 5800 per SDS.* | Link-break test (FRS-DEG-004). | P-022 |
| FRS-NET-008 | **Management plane** | Switch, firewall and hypervisor management interfaces shall be reachable only from Z7 management hosts. | *Management VLAN and local-in policies.* | Management access from other zones refused. | S-504, S-201 |

# 8 System security functions — ISA/IEC 62443-3-3 at SL-T 3

One row per URS-S requirement (FRS-SEC-nnn answers URS-S-nnn). The SR and its SL 3 requirement enhancements are stated in the URS; this section defines how they are configured and tested.

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| **FR 1 — Identification and authentication control** | | | | | |
| FRS-SEC-101 | **SR 1.1 — Human user identification and authentication** | Every interactive interface shall require an individual login. Access arriving from Z9 shall require MFA at the IDMZ jump host. Interfaces that cannot authenticate individuals (chiller TD7 display, drive keypads, meter displays) shall be listed with their physical compensation. | *Desigo CC and Windows hosts: OT domain accounts. Studio 5000: FactoryTalk Security bound to the OT domain. Symbio UI, FC 102 SCF, PowerLogic, QuickServer, S7-1500 (TIA user management): named local accounts. FortiGate and Stratix 5800 admins: TACACS+ or RADIUS. Jump host: MFA. ESXi: domain-joined or vCenter SSO.* | No generic account works on any interface; MFA prompt on the jump host; list of panel-only interfaces matches the physical controls. | S-101, P-030 |
| FRS-SEC-102 | **SR 1.2 — Software process and device identification and authentication** | Devices and processes communicating across conduits shall be authenticated with certificates where both ends support it: OPC UA application certificates (C2, C10), BACnet/SC (C9-n when available, C11, C12), CIP Security (Z2). Each unauthenticated link shall be pinned to listed endpoint addresses. | *Owner PKI issues device and application certificates. FortiGate address objects pin Modbus TCP and BACnet/IP peers (C1-n, C4, C5, C9-n fallback).* | Untrusted certificate refused; traffic from an unlisted address on an unauthenticated conduit blocked and logged. | S-102, Z-005 |
| FRS-SEC-103 | **SR 1.3 — Account management** | Accounts shall be created, changed, disabled and removed centrally where the component supports it; device-local accounts shall be held in the account register and reviewed every **[90 days]**. | *OT domain (two domain controllers, Z7). Local accounts register: Symbio 800, FC 102 SCF, PowerLogic, QuickServer, S7-1500, Desigo PXC, break-glass accounts.* | Register matches every device's account list at SAT; review record exists. | S-103 |
| FRS-SEC-104 | **SR 1.4 — Identifier management** | Identifiers shall be unique per person or device, never re-used within **[24 months]**, and disabled within **[1 business day]** of leaving. | *Naming: first.last for people; role-device for service identities (e.g. svc-kepware-opcua).* | Sample of departed users shows disabled accounts within the limit. | S-104 |
| FRS-SEC-105 | **SR 1.5 — Authenticator management** | All default and factory credentials shall be changed or disabled before SAT. Process credentials shall be held in TPM or HSM where the component supports it. | *Change list: QuickServer label password, PowerLogic defaults, FC 102 legacy 0-6* passwords (SCF enabled), Desigo CC default accounts, switch and firewall bootstrap, S7-1500 access levels set. TPM 2.0 on hosts; Windows LAPS for local administrator.* | Default-credential check per checklist row for every component returns no match. | S-105 |
| FRS-SEC-106 | **SR 1.6 — Wireless access management** | No wireless interface shall be active in Z1–Z10. | *Symbio 800 Air-Fi, Wi-Fi module and cellular uplink not fitted; Bluetooth service app not used; no wireless on switches or firewalls.* | RF survey and inventory show no active wireless. | S-106 |
| FRS-SEC-107 | **SR 1.7 — Strength of password-based authentication** | Password policy shall enforce minimum **[14]** characters, history **[24]**, and lifetime limits for human users; components unable to enforce shall be listed. | *OT domain GPO. Stratix: security passwords min-length and aaa common-criteria policy. FortiGate: password policy. ESXi: Security.PasswordQualityControl. Listed: numeric keypad and display passcodes.* | Weak password rejected on each enforcing component. | S-107 |
| FRS-SEC-108 | **SR 1.8 — Public key infrastructure certificates** | Certificates for HTTPS, OPC UA, BACnet/SC and CIP Security shall come from the owner PKI with defined validity and renewal; self-signed certificates only with written risk acceptance. | *Offline root CA, issuing CA in Z7; renewal alarm at* **[60 days]** *to expiry.* | Certificate inventory: every endpoint shows owner-PKI issuer or a signed risk acceptance. | S-108 |
| FRS-SEC-109 | **SR 1.9 — Strength of public key authentication** | Certificate validation shall check signature, validity, revocation and chain; revocation data shall be published inside OT. | *CRL distribution point in Z7; OPC UA trust lists and rejected-certificate folders reviewed.* | Revoked test certificate refused at FAT. | S-109 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-SEC-110 | **SR 1.10 — Authenticator feedback** | Password entry shall be masked on every interface. | *Native on all listed interfaces.* | Visual check per interface. | S-110 |
| FRS-SEC-111 | **SR 1.11 — Unsuccessful login attempts** | Failed logins shall be limited to **[5]** per **[15 min]** with lockout or delay; operator consoles shall keep a view-only session; break-glass credentials sealed with dual sign-out. | *OT domain lockout policy; FortiGate admin-lockout-threshold and duration; ESXi Security.AccountLockFailures and AccountUnlockTime; QuickServer native lockout.* | Lockout triggers on test account; console stays view-only; break-glass log entry. | S-111 |
| FRS-SEC-112 | **SR 1.12 — System use notification** | A system use banner shall appear before login wherever supported. | *Windows legal notice GPO; Stratix banner login; FortiGate pre-login banner; ESXi login message; PM5560 banner.* | Banner shown on each supporting interface. | S-112 |
| FRS-SEC-113 | **SR 1.13 — Access via untrusted networks** | All access from Z9 shall terminate in Z8; each session shall need explicit approval, MFA, recording and timeout; vendor cloud connectivity shall stay disabled. | *Jump host with approval workflow and session recording; Trane Connect, Siemens cloud connectors and QuickServer cloud registration disabled.* | Direct connection from Z9 to Z1–Z10 impossible; session without approval refused. | S-113, P-021 |
| **FR 2 — Use control** | | | | | |
| FRS-SEC-201 | **SR 2.1 — Authorization enforcement** | Authorization shall follow the role matrix (Viewer, Plant Operator, Shift Supervisor, Plant Engineer, Controls Engineer, Security Administrator, Vendor Service) for people, services and devices, with a time-limited audited supervisor override. | *Desigo CC scope rights per role; FactoryTalk Security permissions; Symbio UI permissions; FC 102 SCF roles; S7-1500 user management roles; OPC UA users: Desigo CC read plus allowlisted writes, Kepware read-only.* | Role test matrix at FAT: each role can do exactly its listed actions. | S-201, P-014 |
| FRS-SEC-202 | **SR 2.2 — Wireless use control** | Unauthorized RF devices in the plant room and panels shall be detected and reported. | *RF survey* **[quarterly]**. | Survey report filed. | S-202 |
| FRS-SEC-203 | **SR 2.3 — Use control for portable and mobile devices** | Only owner-managed laptops and scanned media shall connect; device health shall be checked first. | *Media scanning station in Z7; EWS USB control by GPO; ControlLogix front USB and SD disabled where firmware supports it, otherwise locked panel; Symbio USB service port behind locked door.* | Unscanned media blocked on EWS; panel door-open alarm. | S-203 |
| FRS-SEC-204 | **SR 2.4 — Mobile code** | Mobile code shall run only on managed hosts with integrity checking. | *Windows Defender Application Control (or vendor-qualified allowlisting) on EWS and servers; macros blocked; browser access to device web UIs only from Z6 and Z7 hosts.* | Unsigned script blocked at FAT. | S-204 |
| FRS-SEC-205 | **SR 2.5 — Session lock** | Workstations shall lock after **[15 min]**; operator consoles lock to a view-only state. | *GPO screen lock; Desigo CC operator console configuration.* | Lock observed; alarms still visible on console. | S-205, P-031 |
| FRS-SEC-206 | **SR 2.6 — Remote session termination** | Remote sessions shall end after **[15 min]** idle and be terminable on demand. | *Jump host idle timeout; FortiGate admintimeout; Stratix exec-timeout; ESXi UserVars.ESXiShellInteractiveTimeOut.* | Idle session terminated at limit. | S-206 |
| FRS-SEC-207 | **SR 2.7 — Concurrent session control** | Concurrent sessions shall be limited per account and interface. | *One engineering session per controller; one admin session per FortiGate user.* | Second session refused. | S-207 |
| FRS-SEC-208 | **SR 2.8 — Auditable events** | Security and control events shall be logged with time, source, type, user and outcome and forwarded to the Z7 log collector. | *Desigo CC activity log; Logix controller log and change detection; Symbio 800 audit trail; FC 102 43-6* audit log; S7-1500 diagnostic and security events; FortiGate, Stratix (syslog over TLS where the release supports it), Windows event forwarding, ESXi Syslog.global.logHost.* | Test events from every source arrive at the collector within **[60 s]**. | S-208, S-601 |
| FRS-SEC-209 | **SR 2.9 — Audit storage capacity** | Collector storage shall hold **[12 months]** online and **[3 years]** archive and warn at **[80 %]**. | *Collector sizing per SDS.* | Threshold warning demonstrated. | S-209 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-SEC-210 | **SR 2.10 — Response to audit processing failures** | Audit failures shall alert without stopping control. | *Collector down and source-silent alarms (FRS-HMI-005).* | Collector stopped at FAT: P3 alarm, plant unaffected. | S-210 |
| FRS-SEC-211 | **SR 2.11 — Timestamps** | All components shall synchronize to owner time sources, authenticated where supported. | *Two GPS-disciplined NTP servers in Z7; FortiGate relays per zone; Stratix NTP with authentication keys; ControlLogix via CIP Sync from Stratix 5800 (PTP); Symbio 800, meters, drives and S7-1500 via NTP where supported; others recorded.* | Time offset of every component below **[1 s]** at SAT. | S-211 |
| FRS-SEC-212 | **SR 2.12 — Non-repudiation** | Actions that change the plant or its security shall be attributable to one person. | *Named logins (FRS-SEC-101) plus change comments with MOC number (FRS-LCM-003).* | Audit sample traces each change to one person. | S-212, P-062 |
| **FR 3 — System integrity** | | | | | |
| FRS-SEC-301 | **SR 3.1 — Communication integrity** | Conduit traffic shall be integrity-protected with OPC UA SignAndEncrypt, BACnet/SC, CIP Security, TLS, SSH and SNMPv3 where both ends support them; each remaining insecure protocol shall be confined per URS-Z-005. | *Protocol table (URS 6.4); FRS-INT rows.* | Packet capture per conduit at FAT matches the protocol table. | S-301, Z-005 |
| FRS-SEC-302 | **SR 3.2 — Malicious code protection** | Malicious code protection shall cover hosts and entry points with central reporting. | *Vendor-qualified endpoint protection or allowlisting on Desigo CC, Kepware and EWS hosts; FortiGate AV and IPS at C6 and C7; media scanning station; central console in Z7.* | EICAR test file detected on host and at boundary; event reaches console. | S-302 |
| FRS-SEC-303 | **SR 3.3 — Security functionality verification** | Security functions shall be verifiable automatically against the approved baselines. | *Configuration exports compared with the per-product checklist baselines after each change.* | Compliance report produced at SAT. | S-303, S-706 |
| FRS-SEC-304 | **SR 3.4 — Software and information integrity** | Unauthorized software, firmware, configuration or logic changes shall be detected and notified automatically. | *ControlLogix change detection (Changes to Detect mask, Audit Value) monitored by the plant program and alarmed P3; FC 102 0-91 integrity check enabled, 0-92 hash recorded; Symbio 800 signed firmware; S7-1500 change and download events; host file-integrity monitoring; ESXi Secure Boot and TPM attestation.* | Test logic edit on the FAT system raises an alarm within **[1 min]**. | S-304, P-033 |
| FRS-SEC-305 | **SR 3.5 — Input validation** | Inputs that affect control shall be validated in the controller for range, rate and context. | *AOI per writable tag (FRS-PLC-008).* | Out-of-range and over-rate writes rejected and logged. | S-305, P-012 |
| FRS-SEC-306 | **SR 3.6 — Deterministic output** | Outputs shall go to the defined fail states (FRS-DEG fail-state table). | *Drive communication-loss actions; actuator fail positions; Symbio local default.* | Fail-state test per element. | S-306, P-024 |
| FRS-SEC-307 | **SR 3.7 — Error handling** | Error messages shall not disclose exploitable information. | *Device web UIs and OPC UA servers return generic errors.* | Error-response review at FAT. | S-307 |
| FRS-SEC-308 | **SR 3.8 — Session integrity** | Web and OPC UA sessions shall use unique, unpredictable identifiers invalidated at logout. | *Native on Desigo CC, Symbio UI, FortiGate, PowerLogic web servers and OPC UA servers.* | Session token reuse after logout refused. | S-308 |
| FRS-SEC-309 | **SR 3.9 — Protection of audit information** | Logs shall be protected from change and deletion. | *Forwarding to the collector; collector access for Security Administrator only; archive integrity-protected.* | Non-admin delete attempt refused; archive hash verified. | S-309 |
| **FR 4 — Data confidentiality** | | | | | |
| FRS-SEC-401 | **SR 4.1 — Information confidentiality** | Credentials, project files, backups and network drawings shall be encrypted at rest and in transit across untrusted networks. | *Encrypted backups (Symbio 800, FortiGate); project archives on encrypted volume in Z7; BitLocker on hosts; Kepware historian volume encrypted.* | Backup file unreadable without key; volumes show encryption on. | S-401 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-SEC-402 | **SR 4.2 — Information persistence** | Components leaving service shall be purged or destroyed. | *Decommissioning procedure: ControlLogix destroyed (no documented secure erase); drives, meters and gateways reset then destroyed; host media sanitized.* | Procedure exists and references each component type. | S-402 |
| FRS-SEC-403 | **SR 4.3 — Use of cryptography** | Only current algorithms shall be enabled. | *TLS 1.2 minimum; FortiGate strong-crypto on; OPC UA Basic128Rsa15 and Basic256 disabled; SNMPv3 with AES; SSHv2 only.* | Cipher scan on each TLS endpoint at FAT shows no deprecated suites. | S-403 |
| **FR 5 — Restricted data flow** | | | | | |
| FRS-SEC-501 | **SR 5.1 — Network segmentation** | OT zones shall be physically separate from non-control networks, and the plant shall run with Z8 and Z9 disconnected. | *Dedicated plant fibre and switches; FRS-NET-001.* | SAT: C6 and C7 opened, plant unaffected. | S-501, P-021 |
| FRS-SEC-502 | **SR 5.2 — Zone boundary protection** | Boundaries shall be deny-by-default, support island mode and fail closed without stopping cooling. | *FRS-NET-002, NET-004, NET-005.* | As in those rows. | S-502 |
| FRS-SEC-503 | **SR 5.3 — General purpose person-to-person communication restrictions** | E-mail, chat, social media and file-sharing traffic shall be blocked into and out of OT zones. | *FortiGate application control on all OT conduits.* | Test traffic blocked. | S-503 |
| FRS-SEC-504 | **SR 5.4 — Application partitioning** | Applications and services shall be partitioned by zone and criticality. | *Separate VMs and port groups per zone on ESXi; management only from Z7.* | Port-group review; no cross-zone VM placement. | S-504 |
| **FR 6 — Timely response to events** | | | | | |
| FRS-SEC-601 | **SR 6.1 — Audit log accessibility** | Authorized tools shall read logs programmatically and read-only. | *Collector syslog and API access for the owner monitoring function.* | API read with read-only credential succeeds; write refused. | S-601 |
| FRS-SEC-602 | **SR 6.2 — Continuous monitoring** | Security shall be continuously monitored with alerts to the owner's security operations function. | *Passive OT network monitoring sensors on SPAN ports in Z1–Z5 and Z10 (owner-selected platform); collector correlation rules.* | Simulated scan from Z6 produces an alert. | S-602 |
| **FR 7 — Resource availability** | | | | | |
| FRS-SEC-701 | **SR 7.1 — Denial of service protection** | Supervisory and management traffic shall not starve control traffic; essential functions shall continue under flood conditions. | *Stratix 5800 storm-control and QoS on access ports; FortiGate DoS policies per conduit; chillers continue locally (FRS-CHL-008).* | FAT: broadcast flood on Z2 access port → control I/O unaffected. | S-701, P-022 |
| FRS-SEC-702 | **SR 7.2 — Resource management** | Security functions shall not exhaust control resources. | *Controller OPC UA sessions and subscriptions within limits (FRS-INT-004); endpoint protection exclusions per vendor qualification.* | Controller CPU and memory trends within limits under full load (FRS-PLC-019). | S-702 |
| FRS-SEC-703 | **SR 7.3 — Control system backup** | Backups shall be automated and verified, RPO **[24 h]**. | *Nightly: Symbio 800 automated backup, Desigo CC, FortiGate, Stratix, ESXi host configuration, Kepware project; after every change: Logix .ACD and .L5X, S7-1500 TIA project, MCT 10 archive. Quarterly test restore on the test bench.* | Restore test record; hashes match. | S-703 |
| FRS-SEC-704 | **SR 7.4 — Control system recovery and reconstitution** | The system shall be restorable to a known secure state within the RTO per component class. | *Gold images, verified backups, spares at baseline (FRS-LCM-002), recovery runbooks.* | Timed restore of one controller and one server at SAT. | S-704 |
| FRS-SEC-705 | **SR 7.5 — Emergency power** | Power transfer shall not change the security state. | *Boundary policy and component configurations persist through restart.* | Post-restart configuration matches baseline. | S-705, P-023 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-SEC-706 | **SR 7.6 — Network and security configuration settings** | Components shall match approved baselines and report settings machine-readably. | *Baselines from per-product SL-3 checklists; exported configurations (Logix .L5X, FortiGate .conf, IOS-XE running-config, Desigo CC, Symbio backup, MCT 10) compared at SAT and after change.* | Baseline comparison shows no unapproved difference. | S-706 |
| FRS-SEC-707 | **SR 7.7 — Least functionality** | Unused ports, protocols and services shall be disabled. | *Telnet, HTTP, FTP and SNMPv1/v2c off where possible; FC 102 SNMP (12-83) off if not used; Symbio unused protocols off; ControlLogix web pages off; S7-1500 PUT/GET access left disabled; unused switch ports shut in a parking VLAN.* | Port scan of each component at FAT matches the approved services list. | S-707 |
| FRS-SEC-708 | **SR 7.8 — Control system component inventory** | A complete inventory with versions and SBOMs shall be kept and updated with each change. | *Inventory in Z7; fields include zone, firmware, support end date (FRS-LCM-001).* | Inventory matches site at SAT. | S-708 |

# 9 Component security realization — ISA/IEC 62443-4-2 at SL-C 3

How each component requirement is realized across the worked-example products, with the compensation where a product lacks the capability. Acceptance for every row: the matching row of each applicable per-product SL-3 checklist is adjudicated Pass, or Partial with the listed compensation recorded.

## 9.1 General

| ID | Requirement | Configuration (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|
| FRS-CMP-001 | Each component's SL-C shall be recorded per CR from a feature and configuration mapping, with certificates as supporting evidence. | *Capability statement per product (URS-L-001) checked against the per-product SL-3 checklist.* | Capability register complete for every component before FAT. | C-001 |
| FRS-CMP-002 | Each capability gap shall have a named compensating countermeasure and residual risk accepted by the owner. | *Gap register; compensations allocated in the SDS.* | Every Partial row in the checklists has an entry in the gap register. | C-002 |
| FRS-CMP-003 | Each supplier's secure development lifecycle evidence shall be filed. | *4-1 certificates or equivalent statements.* | File complete at FAT. | C-003 |
| FRS-CMP-004 | Configurations outside a certificate's conditions shall be recorded with their risk impact. | *ControlLogix OPC UA server and redundancy recorded.* | Deviation register entries exist. | C-004 |
| FRS-CMP-005 | Each component shall be classified EDR, HDR, NDR or SAR; software and host recorded separately. | *Inventory component-type field.* | Inventory classification matches the checklist used. | C-005 |

## 9.2 Component requirements common to all component types

| ID | Requirement | Realization (worked example) | Gap and compensation | URS |
|---|---|---|---|---|
| FRS-CMP-101 | **CR 1.1 — Human user identification and authentication** | *Named accounts on Desigo CC, Windows, FactoryTalk, Symbio UI, FC 102 SCF, PowerLogic, QuickServer, S7-1500, FortiGate, Stratix, ESXi.* | *Panel-only interfaces (TD7, keypads, meter displays) → physical access control (FRS-SEC-101).* | C-101 |
| FRS-CMP-102 | **CR 1.2 — Software process and device identification and authentication** | *OPC UA application certificates; BACnet/SC certificates; CIP Security device identity (755T).* | *Modbus TCP, BACnet/IP, EtherNet/IP without CIP Security → endpoint pinning (FRS-SEC-102).* | C-102 |
| FRS-CMP-103 | **CR 1.3 — Account management** | *Domain integration where supported; local account management elsewhere.* | *Local-only devices → account register (FRS-SEC-103).* | C-103 |

| ID | Requirement | Realization (worked example) | Gap and compensation | URS |
|---|---|---|---|---|
| FRS-CMP-104 | **CR 1.4 — Identifier management** | *Unique identifiers per person and service.* | — | C-104 |
| FRS-CMP-105 | **CR 1.5 — Authenticator management** | *All defaults changed; TPM-backed credentials on hosts.* | *Devices without hardware key storage → recorded gap; conduit restriction.* | C-105 |
| FRS-CMP-107 | **CR 1.7 — Strength of password-based authentication** | *Policy enforced on domain, FortiGate, Stratix, ESXi, QuickServer.* | *Numeric keypad codes → physical control.* | C-107 |
| FRS-CMP-108 | **CR 1.8 — Public key infrastructure certificates** | *Owner-PKI certificates on all TLS, OPC UA, BACnet/SC and CIP Security endpoints.* | *Devices that cannot import owner certificates → risk acceptance (FRS-SEC-108).* | C-108 |
| FRS-CMP-109 | **CR 1.9 — Strength of public key-based authentication** | *Chain and revocation checking in OPC UA, BACnet/SC and HTTPS clients.* | *Hardware key protection absent on some devices → recorded gap.* | C-109 |
| FRS-CMP-110 | **CR 1.10 — Authenticator feedback** | *Masked entry everywhere.* | — | C-110 |
| FRS-CMP-111 | **CR 1.11 — Unsuccessful login attempts** | *Lockout or delay on domain, FortiGate, ESXi, QuickServer, FC 102 (0-94).* | *Devices without lockout → rate limiting at FortiGate and monitoring.* | C-111 |
| FRS-CMP-112 | **CR 1.12 — System use notification** | *Banners where supported.* | *Devices without banner → recorded.* | C-112 |
| FRS-CMP-114 | **CR 1.14 — Strength of symmetric key-based authentication** | *Symmetric keys (e.g., NTP authentication keys) stored only on devices and in the Z7 key store.* | — | C-114 |
| FRS-CMP-201 | **CR 2.1 — Authorization enforcement** | *Role-based permissions per FRS-SEC-201.* | *Devices with one shared role → physical and conduit restriction.* | C-201 |
| FRS-CMP-202 | **CR 2.2 — Wireless use control** | *No wireless interfaces active.* | — | C-202 |
| FRS-CMP-205 | **CR 2.5 — Session lock** | *Session lock on hosts and web UIs.* | — | C-205 |
| FRS-CMP-206 | **CR 2.6 — Remote session termination** | *Idle and on-demand termination (FRS-SEC-206).* | — | C-206 |
| FRS-CMP-207 | **CR 2.7 — Concurrent session control** | *Session limits per FRS-SEC-207; controller OPC UA session limits respected.* | — | C-207 |
| FRS-CMP-208 | **CR 2.8 — Auditable events** | *Event logging on every component that supports it, forwarded to Z7.* | *Devices without logs (e.g., MVI56E-MNET) → boundary and network-monitoring logs.* | C-208 |
| FRS-CMP-209 | **CR 2.9 — Audit storage capacity** | *Collector capacity and threshold warning.* | *Device-local log limits recorded.* | C-209 |
| FRS-CMP-210 | **CR 2.10 — Response to audit processing failures** | *Source-silent and collector-down alarms.* | — | C-210 |

| ID | Requirement | Realization (worked example) | Gap and compensation | URS |
|---|---|---|---|---|
| FRS-CMP-211 | **CR 2.11 — Timestamps** | *NTP or PTP on all components that support it.* | *Components without time sync → timestamped by the collector on receipt.* | C-211 |
| FRS-CMP-212 | **CR 2.12 — Non-repudiation** | *Named logins plus MOC comments.* | — | C-212 |
| FRS-CMP-301 | **CR 3.1 — Communication integrity** | *OPC UA SignAndEncrypt, BACnet/SC, CIP Security, TLS, SSH, SNMPv3.* | *Insecure protocols confined per URS-Z-005.* | C-301 |
| FRS-CMP-303 | **CR 3.3 — Security functionality verification** | *Baseline compliance checks after change.* | — | C-303 |
| FRS-CMP-304 | **CR 3.4 — Software and information integrity** | *Change detection, signed firmware, integrity checks and FIM per FRS-SEC-304.* | *Devices without integrity checks → periodic configuration comparison.* | C-304 |
| FRS-CMP-305 | **CR 3.5 — Input validation** | *Controller-side validation of every writable value.* | — | C-305 |
| FRS-CMP-306 | **CR 3.6 — Deterministic output** | *Fail states per FRS-DEG table.* | — | C-306 |
| FRS-CMP-307 | **CR 3.7 — Error handling** | *Generic error responses.* | — | C-307 |
| FRS-CMP-308 | **CR 3.8 — Session integrity** | *Native session handling on web and OPC UA servers.* | — | C-308 |
| FRS-CMP-309 | **CR 3.9 — Protection of audit information** | *Off-device log forwarding; restricted collector access.* | — | C-309 |
| FRS-CMP-401 | **CR 4.1 — Information confidentiality** | *Encrypted backups, volumes and transport.* | *Devices storing credentials in clear (e.g., Config.csv for PM5560) → file handling rules.* | C-401 |
| FRS-CMP-402 | **CR 4.2 — Information persistence** | *Purge or destroy at decommissioning.* | *No secure erase on ControlLogix → destroy.* | C-402 |
| FRS-CMP-403 | **CR 4.3 — Use of cryptography** | *Approved algorithms only (FRS-SEC-403).* | — | C-403 |
| FRS-CMP-501 | **CR 5.1 — Network segmentation** | *Each component placed in exactly one zone; segmentation per FRS-NET-001.* | — | C-501 |
| FRS-CMP-601 | **CR 6.1 — Audit log accessibility** | *Syslog or API access to logs.* | — | C-601 |
| FRS-CMP-602 | **CR 6.2 — Continuous monitoring** | *Passive monitoring of all OT zones.* | — | C-602 |
| FRS-CMP-701 | **CR 7.1 — Denial of service protection** | *Storm control, QoS and DoS policies; local chiller autonomy.* | — | C-701 |
| FRS-CMP-702 | **CR 7.2 — Resource management** | *Resource limits monitored (FRS-PLC-016 to 019).* | — | C-702 |
| FRS-CMP-703 | **CR 7.3 — Control system backup** | *Automated and verified backups per FRS-SEC-703.* | *Devices without automated backup → backup after every change.* | C-703 |

| ID | Requirement | Realization (worked example) | Gap and compensation | URS |
|---|---|---|---|---|
| FRS-CMP-704 | **CR 7.4 — Control system recovery and reconstitution** | *Recovery runbooks and spares.* | — | C-704 |
| FRS-CMP-706 | **CR 7.6 — Network and security configuration settings** | *Baselines from per-product checklists; exported configurations.* | — | C-706 |
| FRS-CMP-707 | **CR 7.7 — Least functionality** | *Unused services disabled per FRS-SEC-707.* | *Services that cannot be disabled (e.g., some meter protocols) → blocked at the boundary.* | C-707 |
| FRS-CMP-708 | **CR 7.8 — Control system component inventory** | *Component identity and version reporting into the inventory.* | — | C-708 |

## 9.3 Component-type-specific requirements

| ID | Requirement | Realization (worked example) | Gap and compensation | URS |
|---|---|---|---|---|
| **EDR** | | | | |
| FRS-DEV-E204 | **EDR 2.4 — Mobile code** | *Symbio UI and PowerLogic web pages used only from Z6 hosts; ControlLogix web pages off.* | — | D-E204 |
| FRS-DEV-E213 | **EDR 2.13 — Use of physical diagnostic and test interfaces** | *Symbio USB service port, drive keypads and USB, ControlLogix USB and SD, meter ports behind locked doors with door-open alarms to the plant controller.* | *Ports that cannot be disabled → physical control and alarm.* | D-E213 |
| FRS-DEV-E302 | **EDR 3.2 — Protection from malicious code** | *Only signed firmware accepted (Symbio 800, ControlLogix, PowerFlex 755T, S7-1500).* | *Devices without signature checks → firmware hash verified before load (URS-L-010).* | D-E302 |
| FRS-DEV-E310 | **EDR 3.10 — Support for updates** | *Firmware updated module by module under MOC with hash verification.* | — | D-E310 |
| FRS-DEV-E311 | **EDR 3.11 — Physical tamper resistance and detection** | *Locked enclosures with door switches; tamper-evident seals on meters.* | — | D-E311 |
| FRS-DEV-E312 | **EDR 3.12 — Provisioning product supplier roots of trust** | *Vendor firmware-signing roots of trust per product (confirm, Annex D).* | — | D-E312 |
| FRS-DEV-E313 | **EDR 3.13 — Provisioning asset owner roots of trust** | *Owner CA loaded for HTTPS, OPC UA and CIP Security where supported.* | *Devices that cannot load owner CA → risk acceptance.* | D-E313 |
| FRS-DEV-E314 | **EDR 3.14 — Integrity of the boot process** | *Boot-time firmware verification where supported.* | *Gaps recorded per product.* | D-E314 |
| **HDR** | | | | |
| FRS-DEV-H204 | **HDR 2.4 — Mobile code** | *Application allowlisting; signed scripts only.* | — | D-H204 |

| ID | Requirement | Realization (worked example) | Gap and compensation | URS |
|---|---|---|---|---|
| FRS-DEV-H213 | **HDR 2.13 — Use of physical diagnostic and test interfaces** | *UEFI password; unused USB disabled; out-of-band management only in Z7.* | — | D-H213 |
| FRS-DEV-H302 | **HDR 3.2 — Protection from malicious code** | *Vendor-qualified endpoint protection or allowlisting.* | — | D-H302 |
| FRS-DEV-H310 | **HDR 3.10 — Support for updates** | *Patches staged in Z8 after vendor qualification; applied host by host.* | — | D-H310 |
| FRS-DEV-H311 | **HDR 3.11 — Physical tamper resistance and detection** | *Locked racks; chassis intrusion alarm where supported.* | — | D-H311 |
| FRS-DEV-H312 | **HDR 3.12 — Provisioning product supplier roots of trust** | *UEFI Secure Boot with vendor keys.* | — | D-H312 |
| FRS-DEV-H313 | **HDR 3.13 — Provisioning asset owner roots of trust** | *Owner certificates in machine stores.* | — | D-H313 |
| FRS-DEV-H314 | **HDR 3.14 — Integrity of the boot process** | *UEFI Secure Boot and TPM measured boot; ESXi attestation.* | — | D-H314 |
| **NDR** | | | | |
| FRS-DEV-N106 | **NDR 1.6 — Wireless access management** | *No wireless on switches or firewalls.* | — | D-N106 |
| FRS-DEV-N113 | **NDR 1.13 — Access via untrusted networks** | *FortiGate IDMZ pair with per-session approval through the jump host.* | — | D-N113 |
| FRS-DEV-N204 | **NDR 2.4 — Mobile code** | *Management web UIs only from Z7.* | — | D-N204 |
| FRS-DEV-N213 | **NDR 2.13 — Use of physical diagnostic and test interfaces** | *Console and USB ports inside locked cabinets.* | — | D-N213 |
| FRS-DEV-N302 | **NDR 3.2 — Protection from malicious code** | *Signed IOS-XE and FortiOS images only.* | — | D-N302 |
| FRS-DEV-N310 | **NDR 3.10 — Support for updates** | *Verified image upgrades, one unit of a redundant pair at a time.* | — | D-N310 |
| FRS-DEV-N311 | **NDR 3.11 — Physical tamper resistance and detection** | *Locked network cabinets with door alarms.* | — | D-N311 |
| FRS-DEV-N312 | **NDR 3.12 — Provisioning product supplier roots of trust** | *Vendor image-signing roots of trust.* | — | D-N312 |

| ID | Requirement | Realization (worked example) | Gap and compensation | URS |
|---|---|---|---|---|
| FRS-DEV-N313 | **NDR 3.13 — Provisioning asset owner roots of trust** | *Owner CA for management HTTPS and RADIUS/TLS.* | — | D-N313 |
| FRS-DEV-N314 | **NDR 3.14 — Integrity of the boot process** | *Image validation at boot.* | — | D-N314 |
| FRS-DEV-N502 | **NDR 5.2 — Zone boundary protection** | *FortiGate deny-by-default, island mode and fail-close (FRS-NET-002, NET-004, NET-005).* | — | D-N502 |
| FRS-DEV-N503 | **NDR 5.3 — General purpose person-to-person communication restrictions** | *FortiGate application control blocks person-to-person protocols.* | — | D-N503 |
| **SAR** | | | | |
| FRS-DEV-S204 | **SAR 2.4 — Mobile code** | *Desigo CC and Kepware clients only on managed hosts.* | — | D-S204 |
| FRS-DEV-S302 | **SAR 3.2 — Protection from malicious code** | *Desigo CC, Kepware and Studio 5000 run with vendor-qualified protection on their hosts.* | — | D-S302 |

# 10 Programming requirements — Secure PLC Coding Top 20

Each of the Top 20 Secure PLC Coding Practices is a programming requirement for the plant controller program. The skid vendor applies the same practices to the S7-1500 program (FRS-PLC-021).

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-PLC-001 | **Practice 1 — Modularize PLC code** | The plant controller program shall implement practice 1 (Modularize PLC code). | *Equipment logic in add-on instructions per equipment type (chiller, pump, fan, valve), each with a test record.* | Code review: no duplicated equipment logic. | S-304, P-010 |
| FRS-PLC-002 | **Practice 2 — Track operating modes** | The plant controller program shall implement practice 2 (Track operating modes). | *Controller mode and keyswitch position read every scan and sent to the BMS; any change alarmed P3.* | Keyswitch change at FAT raises alarm. | S-304, P-033 |
| FRS-PLC-003 | **Practice 3 — Leave operational logic in the PLC wherever feasible** | The plant controller program shall implement practice 3 (Leave operational logic in the PLC wherever feasible). | *All sequencing, staging, failover and validation in the plant controller; the BMS only requests.* | Code and BMS review: no control logic in Desigo CC. | P-012 |
| FRS-PLC-004 | **Practice 4 — Use PLC flags as integrity checks** | The plant controller program shall implement practice 4 (Use PLC flags as integrity checks). | *Heartbeat and sequence-step flags checked for impossible combinations; violation alarmed.* | Forced flag conflict at FAT alarms. | S-304 |
| FRS-PLC-005 | **Practice 5 — Use cryptographic and/or checksum integrity checks for PLC code** | The plant controller program shall implement practice 5 (Use cryptographic and/or checksum integrity checks for PLC code). | *Change detection Audit Value monitored against the approved value; .L5X hash recorded at each release.* | Logic change alarms; hash matches release record. | S-304 |
| FRS-PLC-006 | **Practice 6 — Validate timers and counters** | The plant controller program shall implement practice 6 (Validate timers and counters). | *Timer presets and counter limits checked against allowed ranges each scan; out-of-range forces a safe default and alarms.* | Preset overwritten at FAT → reverted and alarmed. | S-305 |
| FRS-PLC-007 | **Practice 7 — Validate and alert for paired inputs/outputs** | The plant controller program shall implement practice 7 (Validate and alert for paired inputs/outputs). | *Paired signals (valve open/closed limits, pump run command/feedback) checked for impossible states.* | Simulated conflicting limits alarm. | S-305 |
| FRS-PLC-008 | **Practice 8 — Validate HMI input variables at the PLC level, not only at HMI** | The plant controller program shall implement practice 8 (Validate HMI input variables at the PLC level, not only at HMI). | *Every BMS or HMI write passes a validation AOI (range, rate, mode) before use.* | Out-of-range write rejected at the controller. | S-305, P-012 |
| FRS-PLC-009 | **Practice 9 — Validate indirections** | The plant controller program shall implement practice 9 (Validate indirections). | *Array indexes and pointers bounds-checked before use.* | Out-of-bounds index trapped in test. | S-305 |
| FRS-PLC-010 | **Practice 10 — Assign designated register blocks by function (read/write/validate)** | The plant controller program shall implement practice 10 (Assign designated register blocks by function (read/write/validate)). | *BMS interface tags in dedicated UDTs: read-only block, write-request block, validated block; only the write-request block is externally writable.* | OPC UA browse shows only the defined blocks; External Access settings verified. | S-201, S-305 |
| FRS-PLC-011 | **Practice 11 — Instrument for plausibility checks** | The plant controller program shall implement practice 11 (Instrument for plausibility checks). | *Cross-checks: pump running vs DP rise, chiller running vs ΔT, flow vs valve position.* | Implausible combination alarms in simulation. | S-304, P-031 |
| FRS-PLC-012 | **Practice 12 — Validate inputs based on physical plausibility** | The plant controller program shall implement practice 12 (Validate inputs based on physical plausibility). | *Analog inputs checked against physical ranges (e.g., CHW temperature* **[-5 to 40 °C]***) and rate of change; bad quality excluded from control.* | Out-of-range signal flagged and excluded. | S-305 |

| ID | Function | Requirement (testable) | Configuration / programming (worked example) | Acceptance criterion | URS |
|---|---|---|---|---|---|
| FRS-PLC-013 | **Practice 13 — Disable unneeded / unused communication ports and protocols** | The plant controller program shall implement practice 13 (Disable unneeded / unused communication ports and protocols). | *Unused controller and module services off; unused Ethernet ports disabled.* | Port scan matches approved list. | S-707 |
| FRS-PLC-014 | **Practice 14 — Restrict third-party data interfaces** | The plant controller program shall implement practice 14 (Restrict third-party data interfaces). | *Only the OPC UA server, MVI56E-MNET and CIP connections listed in FRS-INT; External Access on all other tags set to None or Read Only.* | Tag browse from an unlisted client fails. | S-201, Z-003 |
| FRS-PLC-015 | **Practice 15 — Define a safe process state in case of a PLC restart** | The plant controller program shall implement practice 15 (Define a safe process state in case of a PLC restart). | *Power-up handler restores the pre-event mode from retentive tags and staggers chiller starts (FRS-DEG-005); outputs held until the first full scan validates inputs.* | Controller restart at FAT returns to the defined state. | S-306, P-023 |
| FRS-PLC-016 | **Practice 16 — Summarize PLC cycle times and trend them on the HMI** | The plant controller program shall implement practice 16 (Summarize PLC cycle times and trend them on the HMI). | *Task scan times captured and trended in Desigo CC and Kepware historian.* | Trend present; alarm above **[80 %]** of watchdog. | S-702 |
| FRS-PLC-017 | **Practice 17 — Log PLC uptime and trend it on the HMI** | The plant controller program shall implement practice 17 (Log PLC uptime and trend it on the HMI). | *Controller uptime tag trended; unexpected reset alarmed.* | Restart visible in trend and alarm log. | S-304 |
| FRS-PLC-018 | **Practice 18 — Log PLC hard stops and trend them on the HMI** | The plant controller program shall implement practice 18 (Log PLC hard stops and trend them on the HMI). | *Major faults captured by the fault routine and sent to the BMS and collector.* | Simulated major fault logged. | S-208 |
| FRS-PLC-019 | **Practice 19 — Monitor PLC memory usage and trend it on the HMI** | The plant controller program shall implement practice 19 (Monitor PLC memory usage and trend it on the HMI). | *Memory usage trended; alarm above* **[80 %]**. | Trend and alarm demonstrated. | S-702 |
| FRS-PLC-020 | **Practice 20 — Trap false alerts** | The plant controller program shall implement practice 20 (Trap false alerts). | *Alarm logic uses deadbands, delays and first-out capture to suppress nuisance and spoofed alerts; trapped events logged.* | Chattering input does not flood alarms; events logged. | P-031, S-304 |
| FRS-PLC-021 | **Skid vendor compliance** | The water-treatment skid vendor shall apply the same 20 practices to the S7-1500 program and declare compliance per practice. | | Vendor compliance statement reviewed at FAT. | P-012, S-304, S-305 |

# Annex A URS → FRS traceability

Generated from the URS and FRS data. Every functional URS ID traces to at least one FRS row; untraced functional IDs: none.

| URS ID | FRS rows | Disposition |
|---|---|---|
| URS-P-001 | FRS-CHL-003, FRS-CHL-004, FRS-HYD-003 | Functional: traced |
| URS-P-002 | FRS-CHL-006, FRS-MOD-003, FRS-DEG-003 | Functional: traced |
| URS-P-003 | FRS-CHL-001, FRS-ECO-003 | Functional: traced |
| URS-P-004 | FRS-TES-001, FRS-TES-002, FRS-TES-003 | Functional: traced |
| URS-P-005 | FRS-ECO-001, FRS-ECO-002, FRS-ECO-003 | Functional: traced |
| URS-P-010 | FRS-CHL-001, FRS-CHL-002, FRS-CHL-003, FRS-CHL-004, FRS-CHL-005, FRS-HYD-001, FRS-HYD-002, FRS-HYD-003, FRS-HYD-005, FRS-CT-001, FRS-CT-002, FRS-CT-004, FRS-ECO-002, FRS-MET-002, FRS-PLC-001 | Functional: traced |
| URS-P-011 | FRS-CHL-006, FRS-HYD-004, FRS-MOD-002 | Functional: traced |
| URS-P-012 | FRS-CHL-007, FRS-WT-001, FRS-HMI-002, FRS-INT-001, FRS-INT-002, FRS-INT-003, FRS-INT-006, FRS-SEC-305, FRS-PLC-003, FRS-PLC-008, FRS-PLC-021 | Functional: traced |
| URS-P-013 | FRS-CHL-008, FRS-DEG-003 | Functional: traced |
| URS-P-014 | FRS-MOD-001, FRS-MOD-002, FRS-MOD-003, FRS-MOD-004, FRS-SEC-201 | Functional: traced |
| URS-P-015 | FRS-CHL-002, FRS-CT-002, FRS-CT-003, FRS-SAF-006, FRS-INT-012 | Functional: traced |
| URS-P-020 | FRS-DEG-001, FRS-HMI-001, FRS-NET-004 | Functional: traced |
| URS-P-021 | FRS-DEG-002, FRS-INT-011, FRS-SEC-113, FRS-SEC-501 | Functional: traced |
| URS-P-022 | FRS-DEG-004, FRS-NET-007, FRS-SEC-701 | Functional: traced |
| URS-P-023 | FRS-DEG-005, FRS-SEC-705, FRS-PLC-015 | Functional: traced |
| URS-P-024 | FRS-HYD-002, FRS-DEG-006, FRS-SEC-306 | Functional: traced |
| URS-P-030 | FRS-HMI-001, FRS-SEC-101 | Functional: traced |
| URS-P-031 | FRS-TES-003, FRS-WT-002, FRS-PRH-001, FRS-HMI-002, FRS-HMI-003, FRS-HMI-004, FRS-INT-005, FRS-SEC-205, FRS-PLC-011, FRS-PLC-020 | Functional: traced |
| URS-P-032 | FRS-HMI-006, FRS-HMI-007, FRS-INT-004 | Functional: traced |
| URS-P-033 | FRS-HMI-005, FRS-SEC-304, FRS-PLC-002 | Functional: traced |
| URS-P-040 | FRS-MET-001, FRS-MET-003, FRS-HMI-008, FRS-INT-009 | Functional: traced |
| URS-P-041 | FRS-MET-002 | Functional: traced |
| URS-P-050 | FRS-PRH-001, FRS-SAF-001, FRS-INT-007, FRS-INT-012 | Functional: traced |
| URS-P-051 | FRS-SAF-002 | Functional: traced |
| URS-P-052 | FRS-SAF-003, FRS-INT-012 | Functional: traced |
| URS-P-053 | FRS-CT-003, FRS-SAF-004 | Functional: traced |
| URS-P-054 | FRS-SAF-005 | Functional: traced |
| URS-P-060 | FRS-LCM-001 | Functional: traced |
| URS-P-061 | FRS-LCM-002 | Functional: traced |
| URS-P-062 | FRS-LCM-003, FRS-SEC-212 | Functional: traced |
| URS-P-063 | FRS-LCM-004 | Functional: traced |
| URS-Z-001 | FRS-NET-001 | Functional: traced |
| URS-Z-002 | FRS-NET-001 | Functional: traced |
| URS-Z-003 | FRS-INT-001, FRS-INT-006, FRS-INT-009, FRS-NET-002, FRS-PLC-014 | Functional: traced |
| URS-Z-004 | FRS-SAF-001, FRS-SAF-003, FRS-SAF-006, FRS-INT-012 | Functional: traced |
| URS-Z-005 | FRS-INT-002, FRS-INT-003, FRS-INT-005, FRS-INT-007, FRS-INT-008, FRS-INT-010, FRS-NET-003, FRS-NET-006, FRS-SEC-102, FRS-SEC-301 | Functional: traced |
| URS-S-101 | FRS-SEC-101 | Functional: traced |
| URS-S-102 | FRS-INT-001, FRS-INT-003, FRS-INT-009, FRS-INT-010, FRS-SEC-102 | Functional: traced |
| URS-S-103 | FRS-SEC-103 | Functional: traced |

| URS ID | FRS rows | Disposition |
|---|---|---|
| URS-S-104 | FRS-SEC-104 | Functional: traced |
| URS-S-105 | FRS-SEC-105 | Functional: traced |
| URS-S-106 | FRS-SEC-106 | Functional: traced |
| URS-S-107 | FRS-SEC-107 | Functional: traced |
| URS-S-108 | FRS-SEC-108 | Functional: traced |
| URS-S-109 | FRS-SEC-109 | Functional: traced |
| URS-S-110 | FRS-SEC-110 | Functional: traced |
| URS-S-111 | FRS-SEC-111 | Functional: traced |
| URS-S-112 | FRS-SEC-112 | Functional: traced |
| URS-S-113 | FRS-INT-011, FRS-SEC-113 | Functional: traced |
| URS-S-201 | FRS-CHL-005, FRS-WT-001, FRS-MOD-001, FRS-LCM-004, FRS-HMI-004, FRS-INT-004, FRS-NET-008, FRS-SEC-201, FRS-PLC-010, FRS-PLC-014 | Functional: traced |
| URS-S-202 | FRS-SEC-202 | Functional: traced |
| URS-S-203 | FRS-SEC-203 | Functional: traced |
| URS-S-204 | FRS-SEC-204 | Functional: traced |
| URS-S-205 | FRS-SEC-205 | Functional: traced |
| URS-S-206 | FRS-SEC-206 | Functional: traced |
| URS-S-207 | FRS-SEC-207 | Functional: traced |
| URS-S-208 | FRS-SEC-208, FRS-PLC-018 | Functional: traced |
| URS-S-209 | FRS-SEC-209 | Functional: traced |
| URS-S-210 | FRS-HMI-005, FRS-SEC-210 | Functional: traced |
| URS-S-211 | FRS-SEC-211 | Functional: traced |
| URS-S-212 | FRS-MOD-001, FRS-LCM-003, FRS-SEC-212 | Functional: traced |
| URS-S-301 | FRS-WT-002, FRS-MET-001, FRS-INT-002, FRS-INT-003, FRS-INT-005, FRS-INT-007, FRS-INT-008, FRS-INT-010, FRS-NET-006, FRS-SEC-301 | Functional: traced |
| URS-S-302 | FRS-SEC-302 | Functional: traced |
| URS-S-303 | FRS-SEC-303 | Functional: traced |
| URS-S-304 | FRS-SAF-004, FRS-HMI-004, FRS-HMI-005, FRS-SEC-304, FRS-PLC-001, FRS-PLC-002, FRS-PLC-004, FRS-PLC-005, FRS-PLC-011, FRS-PLC-017, FRS-PLC-020, FRS-PLC-021 | Functional: traced |
| URS-S-305 | FRS-CHL-001, FRS-CHL-007, FRS-WT-001, FRS-INT-001, FRS-NET-003, FRS-SEC-305, FRS-PLC-006, FRS-PLC-007, FRS-PLC-008, FRS-PLC-009, FRS-PLC-010, FRS-PLC-012, FRS-PLC-021 | Functional: traced |
| URS-S-306 | FRS-CHL-008, FRS-TES-001, FRS-DEG-001, FRS-DEG-003, FRS-DEG-006, FRS-SEC-306, FRS-PLC-015 | Functional: traced |
| URS-S-307 | FRS-SEC-307 | Functional: traced |
| URS-S-308 | FRS-SEC-308 | Functional: traced |
| URS-S-309 | FRS-SEC-309 | Functional: traced |
| URS-S-401 | FRS-HMI-007, FRS-SEC-401 | Functional: traced |
| URS-S-402 | FRS-SEC-402 | Functional: traced |
| URS-S-403 | FRS-INT-003, FRS-SEC-403 | Functional: traced |
| URS-S-501 | FRS-DEG-002, FRS-HMI-007, FRS-NET-001, FRS-SEC-501 | Functional: traced |
| URS-S-502 | FRS-INT-011, FRS-NET-002, FRS-NET-003, FRS-NET-004, FRS-NET-005, FRS-SEC-502 | Functional: traced |
| URS-S-503 | FRS-SEC-503 | Functional: traced |
| URS-S-504 | FRS-NET-008, FRS-SEC-504 | Functional: traced |
| URS-S-601 | FRS-HMI-008, FRS-SEC-208, FRS-SEC-601 | Functional: traced |
| URS-S-602 | FRS-SEC-602 | Functional: traced |
| URS-S-701 | FRS-CHL-008, FRS-DEG-004, FRS-SEC-701 | Functional: traced |

| URS ID | FRS rows | Disposition |
|---|---|---|
| URS-S-702 | FRS-INT-004, FRS-SEC-702, FRS-PLC-016, FRS-PLC-019 | Functional: traced |
| URS-S-703 | FRS-SEC-703 | Functional: traced |
| URS-S-704 | FRS-SEC-704 | Functional: traced |
| URS-S-705 | FRS-DEG-005, FRS-SEC-705 | Functional: traced |
| URS-S-706 | FRS-SEC-303, FRS-SEC-706 | Functional: traced |
| URS-S-707 | FRS-SEC-707, FRS-PLC-013 | Functional: traced |
| URS-S-708 | FRS-LCM-001, FRS-SEC-708 | Functional: traced |
| URS-C-001 | FRS-CMP-001 | Functional: traced |
| URS-C-002 | FRS-CMP-002 | Functional: traced |
| URS-C-003 | FRS-CMP-003 | Functional: traced |
| URS-C-004 | FRS-CMP-004 | Functional: traced |
| URS-C-005 | FRS-CMP-005 | Functional: traced |
| URS-C-101 | FRS-CMP-101 | Functional: traced |
| URS-C-102 | FRS-CMP-102 | Functional: traced |
| URS-C-103 | FRS-CMP-103 | Functional: traced |
| URS-C-104 | FRS-CMP-104 | Functional: traced |
| URS-C-105 | FRS-CMP-105 | Functional: traced |
| URS-C-107 | FRS-CMP-107 | Functional: traced |
| URS-C-108 | FRS-CMP-108 | Functional: traced |
| URS-C-109 | FRS-CMP-109 | Functional: traced |
| URS-C-110 | FRS-CMP-110 | Functional: traced |
| URS-C-111 | FRS-CMP-111 | Functional: traced |
| URS-C-112 | FRS-CMP-112 | Functional: traced |
| URS-C-114 | FRS-CMP-114 | Functional: traced |
| URS-C-201 | FRS-CMP-201 | Functional: traced |
| URS-C-202 | FRS-CMP-202 | Functional: traced |
| URS-C-205 | FRS-CMP-205 | Functional: traced |
| URS-C-206 | FRS-CMP-206 | Functional: traced |
| URS-C-207 | FRS-CMP-207 | Functional: traced |
| URS-C-208 | FRS-CMP-208 | Functional: traced |
| URS-C-209 | FRS-CMP-209 | Functional: traced |
| URS-C-210 | FRS-CMP-210 | Functional: traced |
| URS-C-211 | FRS-CMP-211 | Functional: traced |
| URS-C-212 | FRS-CMP-212 | Functional: traced |
| URS-C-301 | FRS-CMP-301 | Functional: traced |
| URS-C-303 | FRS-CMP-303 | Functional: traced |
| URS-C-304 | FRS-CMP-304 | Functional: traced |
| URS-C-305 | FRS-CMP-305 | Functional: traced |
| URS-C-306 | FRS-CMP-306 | Functional: traced |
| URS-C-307 | FRS-CMP-307 | Functional: traced |
| URS-C-308 | FRS-CMP-308 | Functional: traced |
| URS-C-309 | FRS-CMP-309 | Functional: traced |
| URS-C-401 | FRS-CMP-401 | Functional: traced |
| URS-C-402 | FRS-CMP-402 | Functional: traced |
| URS-C-403 | FRS-CMP-403 | Functional: traced |

| URS ID | FRS rows | Disposition |
|---|---|---|
| URS-C-501 | FRS-CMP-501 | Functional: traced |
| URS-C-601 | FRS-CMP-601 | Functional: traced |
| URS-C-602 | FRS-CMP-602 | Functional: traced |
| URS-C-701 | FRS-CMP-701 | Functional: traced |
| URS-C-702 | FRS-CMP-702 | Functional: traced |
| URS-C-703 | FRS-CMP-703 | Functional: traced |
| URS-C-704 | FRS-CMP-704 | Functional: traced |
| URS-C-706 | FRS-CMP-706 | Functional: traced |
| URS-C-707 | FRS-CMP-707 | Functional: traced |
| URS-C-708 | FRS-CMP-708 | Functional: traced |
| URS-D-E204 | FRS-DEV-E204 | Functional: traced |
| URS-D-E213 | FRS-DEV-E213 | Functional: traced |
| URS-D-E302 | FRS-DEV-E302 | Functional: traced |
| URS-D-E310 | FRS-DEV-E310 | Functional: traced |
| URS-D-E311 | FRS-DEV-E311 | Functional: traced |
| URS-D-E312 | FRS-DEV-E312 | Functional: traced |
| URS-D-E313 | FRS-DEV-E313 | Functional: traced |
| URS-D-E314 | FRS-DEV-E314 | Functional: traced |
| URS-D-H204 | FRS-DEV-H204 | Functional: traced |
| URS-D-H213 | FRS-DEV-H213 | Functional: traced |
| URS-D-H302 | FRS-DEV-H302 | Functional: traced |
| URS-D-H310 | FRS-DEV-H310 | Functional: traced |
| URS-D-H311 | FRS-DEV-H311 | Functional: traced |
| URS-D-H312 | FRS-DEV-H312 | Functional: traced |
| URS-D-H313 | FRS-DEV-H313 | Functional: traced |
| URS-D-H314 | FRS-DEV-H314 | Functional: traced |
| URS-D-N106 | FRS-DEV-N106 | Functional: traced |
| URS-D-N113 | FRS-DEV-N113 | Functional: traced |
| URS-D-N204 | FRS-DEV-N204 | Functional: traced |
| URS-D-N213 | FRS-DEV-N213 | Functional: traced |
| URS-D-N302 | FRS-DEV-N302 | Functional: traced |
| URS-D-N310 | FRS-DEV-N310 | Functional: traced |
| URS-D-N311 | FRS-DEV-N311 | Functional: traced |
| URS-D-N312 | FRS-DEV-N312 | Functional: traced |
| URS-D-N313 | FRS-DEV-N313 | Functional: traced |
| URS-D-N314 | FRS-DEV-N314 | Functional: traced |
| URS-D-N502 | FRS-DEV-N502 | Functional: traced |
| URS-D-N503 | FRS-DEV-N503 | Functional: traced |
| URS-D-S204 | FRS-DEV-S204 | Functional: traced |
| URS-D-S302 | FRS-DEV-S302 | Functional: traced |
| URS-L-001 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-002 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-003 | — | Non-functional: carried to SDS, HDS and V&V work instruction |

| URS ID | FRS rows | Disposition |
|---|---|---|
| URS-L-004 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-005 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-006 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-007 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-008 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-009 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-010 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-011 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-012 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-013 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-014 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-015 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-L-016 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-001 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-002 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-003 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-004 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-005 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-006 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-007 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-008 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-009 | — | Non-functional: carried to SDS, HDS and V&V work instruction |
| URS-V-010 | — | Non-functional: carried to SDS, HDS and V&V work instruction |

# Annex B  Requirement count

| Area | Rows |
|---|---|
| Process and operational (CHL, HYD, CT, ECO, TES, WT, PRH, MET, MOD, DEG, SAF, LCM) | 49 |
| Operator interface and historian (HMI) | 8 |
| Interfaces (INT) | 12 |
| Zones and conduits (NET) | 8 |
| System security (SEC) | 51 |

| Area | Rows |
|---|---|
| Component realization (CMP) | 48 |
| Component-type realization (DEV) | 30 |
| Programming (PLC) | 21 |
| Total FRS rows | 227 |

## Annex C  FRS → test traceability

Completed with the SDS, HDS and the V&V work instruction.

| FRS ID | SDS / HDS | FAT test | SAT / V&V test | Status |
|---|---|---|---|---|
| FRS-CHL-006 | [SDS-4.2] | [FAT-CHL-06] | [SAT-CHL-06] | [Open] |
| FRS-SEC-102 | [SDS-6.3] | [FAT-SEC-12] | [VV-EDR-1.2] | [Open] |
| | | | | |
| | | | | |

## Annex D  Open items to confirm

Carried from URS Annex D (OI-01 to OI-11) plus the items below.

| ID | Party | Item |
|---|---|---|
| OI-12 | Trane | Symbio 800 Modbus TCP register map for enable, setpoint and demand limit, and the setting that defines behaviour on loss of BAS communication (FRS-CHL-008, INT-001). |
| OI-13 | Rockwell | PowerFlex 755T communication-loss action parameter on the EtherNet/IP port; ControlLogix 5580 options to disable front USB and SD on the chosen firmware. |
| OI-14 | Fortinet | FortiGate industrial signature coverage for Modbus function-code and BACnet write-service filtering (FRS-NET-003, INT-002). |
| OI-15 | Siemens | S7-1500 method to expose a read-only Modbus server data area to the plant controller (FRS-WT-001). |
| OI-16 | Danfoss / Endress+Hauser | CIP Security support on FC 102 MCA 121 and Proline 300/500 EtherNet/IP (FRS-INT-010). |
| OI-17 | PTC (Kepware) | Kepware historian sizing for [7 years] at the listed sample rates, encryption at rest, and outbound forwarding method to campus aggregation. |
| OI-18 | All suppliers | Firmware-signing roots of trust and boot-integrity behaviour per product (FRS-DEV-E312, E314). |